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Magnetostratigraphy of Neogene Andean foreland-basin strata, lat 33°S, Mendoza Province, Argentina

M.V. Irigoyen* *Department of Earth Sciences, Carleton University and Ottawa-Carleton Geoscience Centre, Ottawa, Ontario K1S 5B6, Canada*

K.L. Buchan† *Geological Survey of Canada, 601 Booth Street, Ottawa, Ontario K1A 0E8, Canada*

R.L. Brown *Department of Earth Sciences, Carleton University and Ottawa-Carleton Geoscience Centre, Ottawa, Ontario K1S 5B6, Canada*

ABSTRACT

Neogene nonmarine strata in the southernmost part of the Precordillera thrust belt of the Andes of Argentina accumulated during the Miocene and Pliocene in response to eastward advance of thrust sheets located to the west. To link the sequence of deformational events with the sedimentary record, five stratigraphic sections in the Mendoza Province were dated using magnetic polarity stratigraphy calibrated with ^{40}Ar-^{39}Ar dates from interbedded tephras. These data establish age control on tectonic activity, volcanism, and deposition.

The existence of a backarc foreland basin that developed prior to or ca. 16 Ma is recorded by the oldest strata, the ca. 15.7–12.2 Ma (Baksi time scale) Mariño Formation. Deposition of this unit is linked to a major phase of thrust activity in the central part of the Aconcagua fold-and-thrust belt, in the Principal Cordillera. Initiation of deposition coincides with an eastward shift of the subduction-related magmatic foci to the Argentine Principal Cordillera. The overlying ca. 11.7–9.0 Ma La Pilona Formation is thought to record the initiation of exhumation of the Frontal Cordillera to the northwest of the study region. An extensive ash-rich unit, the 8.9 (locally 9.5?) to 8.7 Ma Tobas Angostura Formation, is correlated with the latest and probably the largest extrusive episode in the Principal Cordillera. Chronologic data constrain the timing of oil generation in the Cuyo basin and development of prospective closures. The new data also require that land mammalian fossils found in the Río de los Pozos Formation are late Miocene, rather than Pliocene, as previously believed.

Keywords: Andes, Ar/Ar, Argentina, magnetostratigraphy, Miocene, Precordillera.

INTRODUCTION

In the late 1950s, it was proposed that the present-day configuration of the Andes Mountains between lat 33° and 34°S was the result of several pulses of uplift and deformation of tectonically active belts, the locus of deformation migrating eastward beginning in the Neogene (cf. Polanski, 1958, 1963). Incorporating this hypothesis within a plate tectonic framework, Ramos (1988) described a tectonic model for the Neogene evolution of the Andes and linked the Cenozoic synorogenic deposits that crop out on the eastern flank of the Andes with the eastward migration of the orogenic thrust front during the late Tertiary. The chronology of thrusting required testing, however, because of a lack of crosscutting relationships and geochronologic data. Because magnetostratigraphic studies of the sedimentary record in other foreland basins of Argentina (e.g., Johnson et al., 1986; Bercowski et al., 1986; Jordan et al., 1988, 1990, 1996; Reynolds et al., 1990; Milana, 1991) proved to be successful in deciphering the time of motion on thrust faults, a similar approach was attempted in this study of the Tertiary Mendoza strata, near lat 33°S (Fig. 1). The most complete stratigraphic record is exposed in the south-plunging La Pilona anticline (Fig. 2A) where most of the sampling was carried out. Additional sampling was conducted on the eastern flank of the Tupungato anticline, located ~25 km farther south (Fig. 2B).

GEOLOGICAL SETTING

The study area (~33°S) is situated above a dip transition in the subducting Nazca plate. To the north, the underthrusting plate is nearly horizontal at depth and, to the south of 33°S, it assumes a dip of ~30° (e.g., Bevis and Isacks, 1984). Changes in large-scale tectonic features of the orogen are associated with this change in the subducting plate geometry. The flat-subduction segment of Chile and Argentina (28°–33°S) is characterized by the lack of an active magmatic arc, the development of the Precordillera as a thin-skinned thrust belt, and the basement block uplifts of the Sierras Pampeanas to the east (Jordan et al., 1983). In contrast, the Andes to the south of 33°15′S are characterized by an active magmatic arc, extensive backarc volcanism, and a narrower zone of foreland deformation (Jordan et al., 1983; Ramos and Nullo, 1993; Bermúdez et al., 1993).

At lat 33°S, the Principal and Frontal Cordilleras and the Precordillera of the Argentine Andes (Fig. 1) plus the Sierras Pampeanas to the east represent four north-trending morphotectonic provinces differentiated on the basis of structural styles, geologic evolution, and morphologic expression. The westernmost of these, the Principal Cordillera or Main Andes, is composed primarily of a thick sequence of Mesozoic marine strata that interfingers with a thick accumulation of Upper Jurassic–Lower Cretaceous volcanic rocks (Yrigoyen, 1979). These strata, which are overlain by Miocene volcanic and pyroclastic rocks (Complejo Volcánico Aconcagua; Ramos et al., 1996), display a thin-skinned deformation and form the early to late Miocene

*Present address: Geological and Mining Survey of Argentina, Julio A. Roca 651, Piso 10 (1322), Buenos Aires, Argentina; e-mail: mirigo@secind.mecon.gov.ar.
†E-mail: kbuchan@nrcan.gs.ca.

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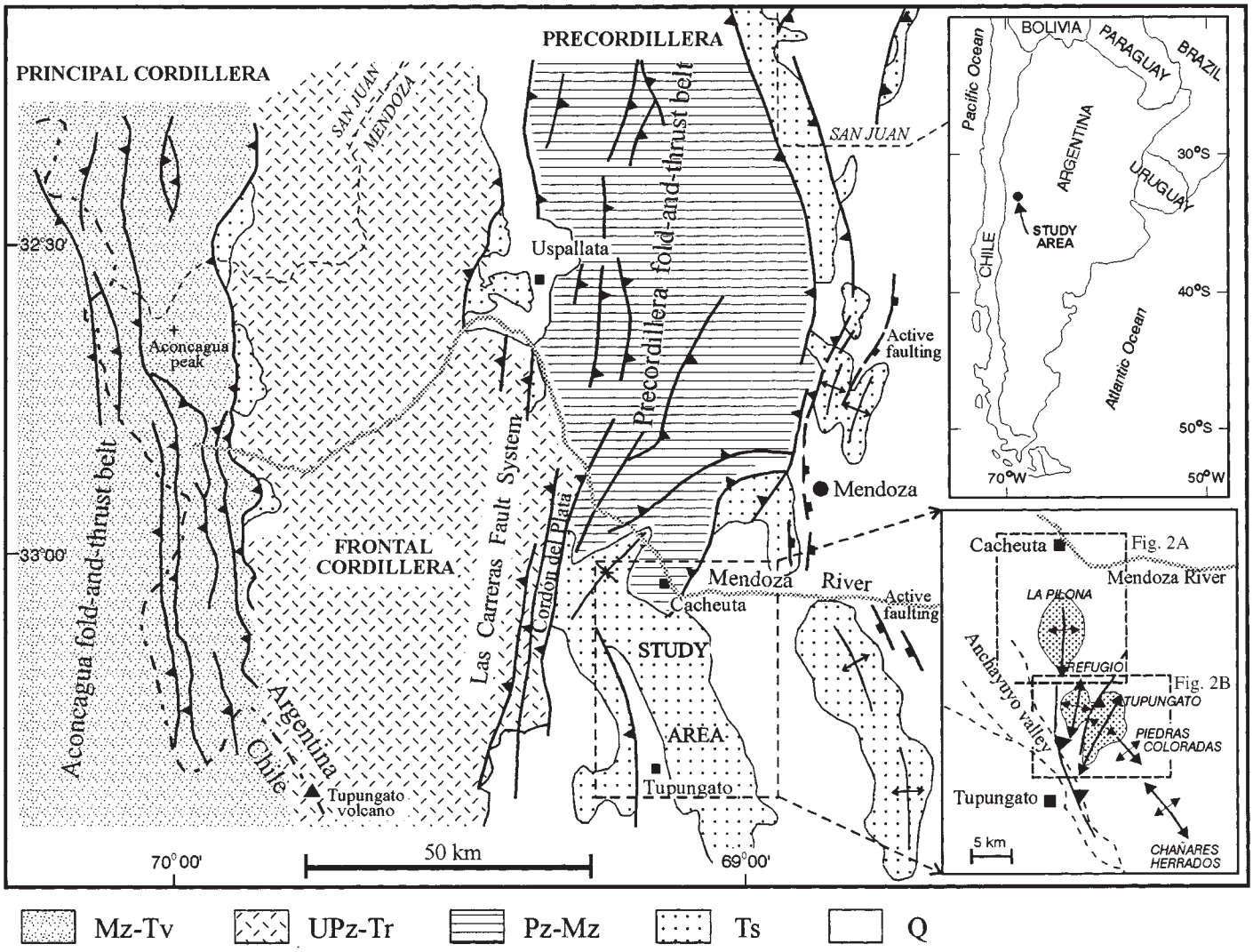


Figure 1. Simplified regional map (modified from Kozlowski et al., 1993) showing the principal tectonic elements associated with the Neogene Mendoza strata of western Argentina. The bedrock geology of the morphotectonic provinces consists primarily of: Mz-Tv—Mesozoic marine sedimentary rocks and Mesozoic-Tertiary volcanic rocks; Upz-Tr—Carboniferous to Triassic silicic and andesitic magmatic rocks and upper Paleozoic sedimentary rocks; Pz-Mz—Paleozoic marine strata and upper Paleozoic–Triassic nonmarine clastic strata; Ts—Tertiary nonmarine strata; and Q—Quaternary cover.

Aconcagua fold-and-thrust belt (Ramos, 1988; Mpodozis and Ramos, 1990). To the east, the Frontal Cordillera mainly exposes an assemblage of Paleozoic andesitic to silicic magmatic and sedimentary rocks (Caminos, 1979). During the Andean orogeny, this morphotectonic unit apparently behaved as a rigid block, disrupted by high-angle reverse faults, but probably not recording significant internal shortening at upper crustal levels (Mpodozis and Ramos, 1990; Allmendinger et al., 1990). An imbricate structure known as Las Carreras fault system (see Fig. 1) forms the eastern boundary of the Frontal Cordillera. The Precordillera is a predominantly east-verging fold-and-thrust belt between 28° and 33°S, with a thin-skinned structure that has been developed on early to late Paleozoic sedimentary rocks (Baldis and Chebli, 1969). Deformation of the Precordillera to the north of the study area (~31°S) started ca. 20 Ma and continued to the present (Jordan et al., 1993a, 1993b).

The study area is located at the southernmost portion of the Precordillera, where the thrust belt is expressed at the surface by deformed Tertiary strata (Fig. 1). The oldest rocks in the region are of Ordovician age (Varela et al., 1993; Fig. 2A). Middle to Upper Triassic nonmarine strata and subordinate volcanic rocks overlie volcanic rocks of the Choiyoi Group. Hydrocarbon reserves generated in the Mendoza Triassic rocks are concentrated in several oil fields and are largely controlled by structural traps, such as the La Pilona and Tupungato anticlines, the two study localities (Dellapé and Hegedus, 1993). About 3500 m of deformed Paleogene to Pliocene-Pleistocene(?) nonmarine clastic strata disconformably overlie the Triassic rocks (Rolleri and Fernández Garrasino, 1979). The Cenozoic succession is generally deformed into south-plunging

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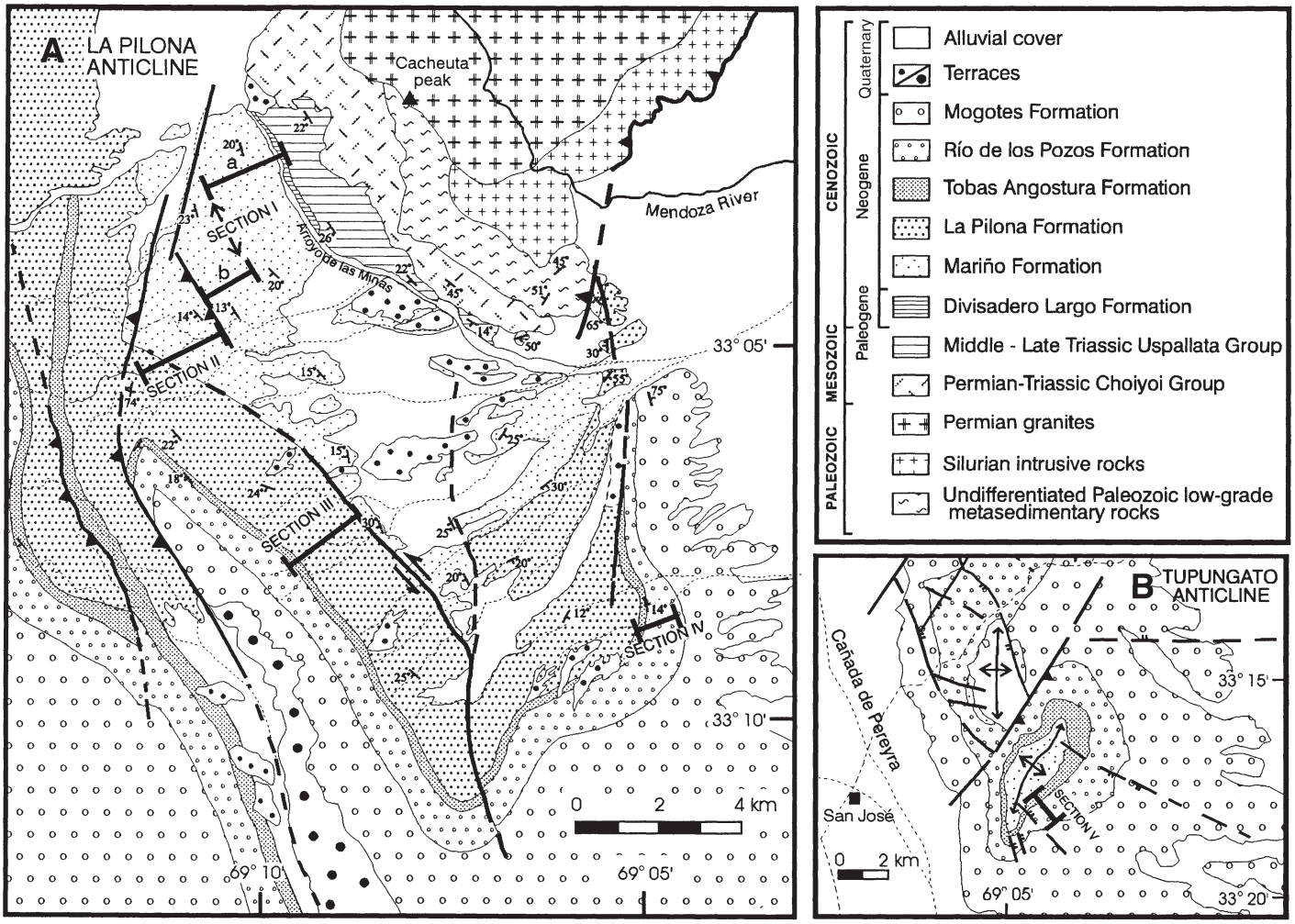


Figure 2. Geology and magnetostratigraphic sections of (A) the La Pilona anticline (sections I-IV) and (B) the Tupungato anticline (section V). Geology is based on a compilation of mapping from this study and previous reports (e.g., Rolleri and De Giusto, 1950; De la Motta, 1957; Varela et al., 1993).

or doubly plunging folds defining two north-northwest–striking anticline systems, the westernmost of which comprises the La Pilona and Tupungato structures (Fig. 2). The eastern anticlinal system marks the modern eastern deformation front.

STRATIGRAPHY OF THE TERTIARY DEPOSITS

The most complete stratigraphic record of the Cenozoic succession is exposed in the La Pilona area (Fig. 2A) and consists of six Tertiary formations. Six stratigraphic horizons consisting of air-fall deposits and reworked tuff beds in the Neogene succession have been dated between 12 and 8 Ma using ^{40}Ar-^{39}Ar geochronology (Irigoyen, 1997; Irigoyen et al. 1999). The ^{40}Ar-^{39}Ar data are summarized in Table 1 (laboratory procedures are given in the Appendix). Formational units are briefly described here in ascending stratigraphic order (following the nomenclature of Yrigoyen, 1993a).

Divisadero Largo Formation

The Divisadero Largo Formation reaches a thickness of ~150 m in the study region but is locally 35 m thick in the Tosca de Mariño section (section I; Fig. 2A). It comprises two members: a basal member of sandstones with gypsum and anhydrite horizons, and an upper member of mostly varicolored mudstones (Yrigoyen, 1993a). Based on detailed facies analysis in northern Mendoza, Kokogian and

TABLE 1. SUMMARY OF ^{40}Ar-^{39}Ar RESULTS

Sample	Unit*	Min†	Method	Age§
Md-5	TA	Hb	Laser step-heating	8.64 ± 0.17
Md-5		Bt	Laser total fusion	8.85 ± 0.37
Md-6	LP	Hb	Laser step-heating	10.42 ± 0.20
Md-6		Bt	Laser total fusion	10.75 ± 0.29
Md-10	LP	Hb	Furnace step-heating	9.30 ± 0.68
Md-10		Bt	Laser total fusion	8.39 ± 0.18
Md-12	MA	Hb	Laser step-heating	12.03 ± 0.45
Md-13	LP	Hb	Laser step-heating	9.63 ± 0.22
Md-15	TA	Hb	Laser step-heating	9.54 ± 0.13
Md-15		Bt	Laser step-heating	8.62 ± 0.95

Note: After Irigoyen et al. (1999)
*TA = Tobas Angostura Formation, LP = La Pilona Formation, MA = Mariño Formation.
†Min (mineral): Hb = hornblende, Bt = biotite.
§Preferred age (as described in Irigoyen et al., 1999). Uncertainties are quoted at the 2σ confidence interval. Ages were calculated using formulas and error assessments detailed in Roddick (1988).

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Mancilla (1989) interpreted this formation as an ancient continental sabkha. The stratigraphic age of this formation is constrained by vertebrate fossils from the upper member and may range from upper Eocene to lower Miocene (e.g., Milana et al., 1993, and references therein).

Mariño Formation

The Mariño Formation, which unconformably overlies the Divisadero Largo Formation (Yrigoyen, 1993a), is subdivided into three members. The lower member attains a thickness of ~50 m in the Tosca de Mariño section (section I; Fig. 2A) and consists of purplish and greenish conglomerate and sandstone beds alternating with red mudstone and a few bentonite horizons. The intermediate member mainly comprises a distinctive 180-m-thick pile of cross-bedded sandstones rich in glass shards. The upper member is ~800 m thick and consists of sandstones and gray conglomerates interbedded with red mudstones. The contacts between the members mark recognized changes of depositional facies. The lower and upper members are interpreted to record fluvial-alluvial deposition in an arid to semiarid setting dominated by ephemeral-stream processes; strata of the intermediate section are of eolian origin (Irigoyen, 1997).

The age of the Mariño Formation was previously poorly constrained, ranging from early Oligocene to middle Miocene (e.g., Yrigoyen, 1993a; Jordan and Ortiz, 1987). A reworked tuff horizon collected from the uppermost part of this unit in the Arroyo del Agua Blanca section (section II; Fig. 2A) was dated as 12.03 ± 0.45 Ma (sample Md-12, Table 1) (Irigoyen, 1997; Irigoyen et al., 1999).

La Pilona Formation

The La Pilona Formation is as much as ~800 m thick in the La Pilona area. Its basal portion includes strata previously termed Tobas Grises Inferiores (Lower gray tuffs; Truempy and Lhez, 1937). Field mapping in this study did not find any basis for distinguishing a separate basal unit. This formation unconformably overlies the Mariño Formation (Yrigoyen, 1993a), although the unconformity is not obvious in the study sections. The lowest third of the La Pilona Formation consists mainly of gray sandstones and conglomerates. It is overlain by a tuffaceous finer grained succession containing a few ash-fall horizons, one of which is known as Horizonte de Ceniza Blanca (White ash horizon; Truempy and Lhez, 1937). The ~200-m-thick uppermost part of this unit mostly comprises conglomerates. La Pilona Formation is interpreted to record fluvial and clay-playa deposition (Irigoyen, 1997).

Based on radiometric data, Yrigoyen (1993a, 1993b) assigned the La Pilona Formation to the late Miocene (ca. 11.1–9.8 Ma). Irigoyen (1997) and Irigoyen et al. (1999) dated two samples collected near the middle of the formation (Md-6, White ash horizon, and Md-13 of Table 1). For sample Md-6, Irigoyen et al. (1999) have taken 10.53 ± 0.16 Ma as the best age estimate, which is a weighted average of 10.42 ± 0.20 Ma from laser step-heating analysis on hornblende and 10.75 ± 0.29 Ma from total fusion analysis on biotite. Sample Md-13 hornblende yielded an age of 9.63 ± 0.22 Ma (Table 1).

Tobas Angostura Formation

The extensive ash-rich Tobas Angostura Formation attains a thickness of 100–120 m in the Tupungato area and overlies the La Pilona Formation with apparent unconformity (Yrigoyen, 1993a). These volcanogenic deposits include coarse- and fine-grained tuffs, reworked tuffs, sandstones, and conglomerates with varying amounts of pyroclastic material, agglomerates, lapilli layers, and air-fall horizons. This formation is interpreted to record the response of a semiarid fluvial system to the pyroclastic volcanism, where deposition occurred in broad extensive areas (Irigoyen, 1997).

Based on radiometric data, Yrigoyen (1993a, 1993b) assigned this formation to the late Miocene (ca. 9.8 Ma). Three samples (Md-5, Md-10, Md-15; Table 1) collected from air-fall ash beds in three different locations of the study area provided age constraints on this unit. Irigoyen (1997) and Irigoyen et al. (1999) reported evidence of loss of radiogenic ^{40}Ar from sample Md-15 biotite in laser step-heating experiments. Therefore, they interpreted the hornblende ages of 8.64 ± 0.17, 9.30 ± 0.68, and 9.54 ± 0.13 Ma from samples Md-5, Md-10, and Md-15, respectively, as the best age estimates.

Río de los Pozos Formation

The Río de los Pozos Formation is gradational from the Tobas Angostura Formation (Yrigoyen, 1993a). In the Tupungato area, this formation exhibits a variable thickness up to 400 m, due to the unconformity with the overlying Mogotes Formation (Yrigoyen, 1993a). The Río de los Pozos Formation comprises mudstones, yellowish tuffaceous mudstones and sandstones, and gray conglomerates. These strata are interpreted to record deposition from unconfined and channelized flows in a semiarid setting (Irigoyen, 1997). Abundant remains of land mammal fossils were collected from this unit in the Tupungato area (cf. Yrigoyen, 1993b). The magnetostratigraphic study proved to be helpful in calibrating the ages of these fossils.

Mogotes Formation

The Mogotes Formation, which was not sampled for magnetostratigraphy, consists of boulder conglomerates interbedded with red mudstones and sandstones and a few tuffaceous horizons. These deposits probably represent proximal alluvial-fan facies, although a possible glacial origin has been suggested (Cortés, *in* Ramos, 1990).

MAGNETOSTRATIGRAPHY

Sampling Procedure

Five profiles (I–V, Figs. 3, 4, locations in Fig. 2) were studied to establish the magnetic polarity stratigraphy of the Neogene units, except for the Mogotes Formation. Sampling followed the procedures established by Johnson et al. (1975). We collected a minimum of three oriented hand samples at each sampling site. Where possible, they were spaced ~1.5 m apart within the same sedimentary bed. We sampled a total of 111 paleomagnetic sites separated by an average of 25 m through the stratigraphy. The preferred lithologies for paleomagnetic sampling were clay and silt, although fine-grained sandstones were collected when finer grained rocks were not available (i.e., in the eolian member of the Mariño Formation) and/or when they were too fissile to be easily sampled. In some instances, the stratigraphic separation between sites was larger than the average due either to lack of exposure (middle part of the La Higuerita section) or to the occurrence of thick conglomerate sequences. Stratigraphic thicknesses were measured with a tape and/or a Jacob staff using a Brunton compass and bench marks. Paleomagnetic laboratory and analytical procedures are described in the Appendix.

Results

The stratigraphy of the five sampling profiles, locations of the paleomagnetic sites, and a summary of the paleomagnetic data are shown in Figures 3 and 4. Most samples are stably magnetized. Typical examples of thermal demagnetization are shown in Figure 5. The natural remanent magnetization (NRM) of most of the samples includes two components of magnetization. The characteristic component has either a north and up or south and down direction and blocking temperatures that are typically above 250 °C. It is interpreted as a detrital remanent magnetization (DRM) carried by titanomagnetite and/or titanohematite. The

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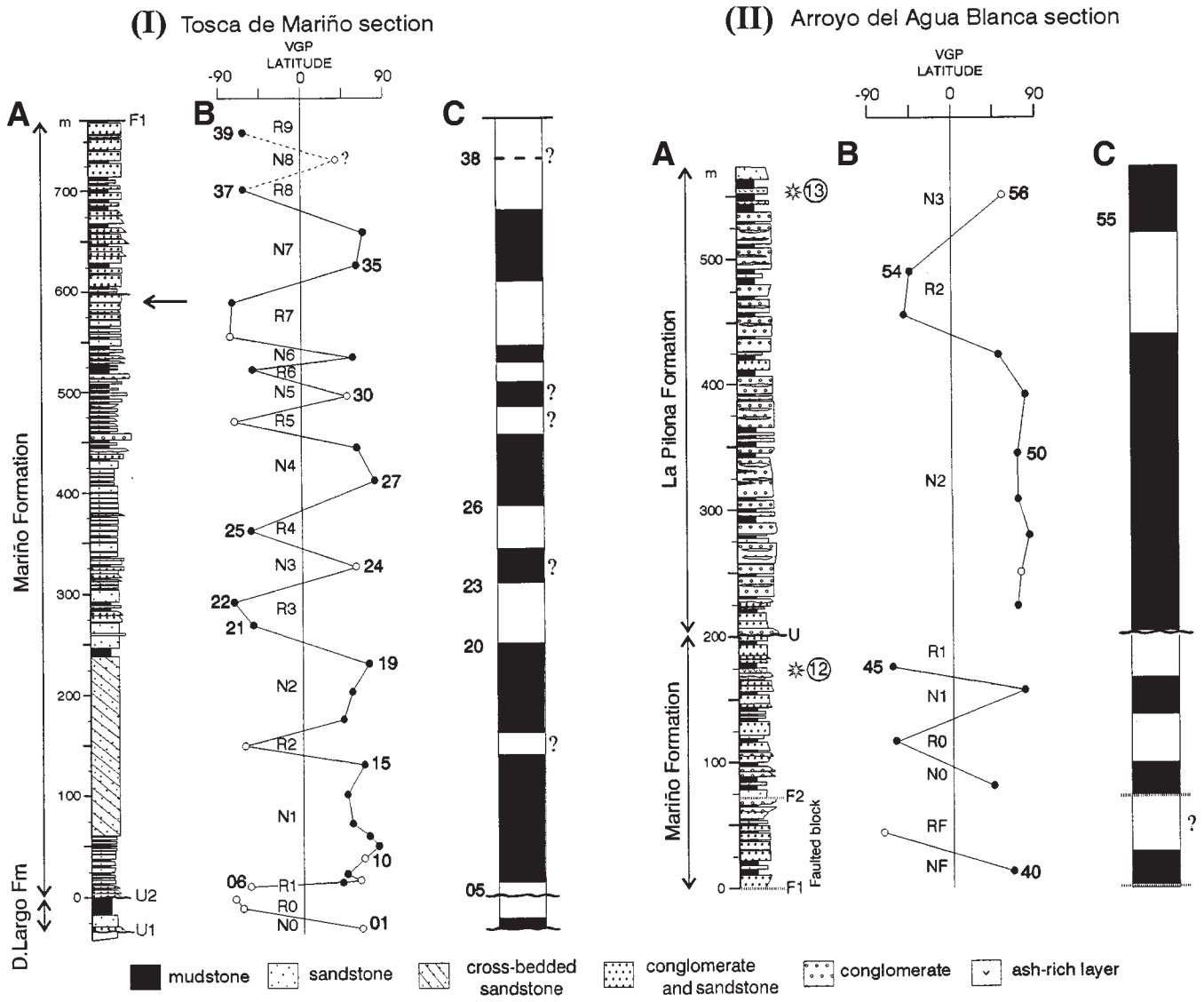


Figure 3. Magnetic polarity stratigraphy of the Tosca de Mariño (I) and Arroyo del Agua Blanca (II) sections. (A) Simplified stratigraphic section showing formations, basic textures, faults (F), unconformities (U), and dated tuff horizons (stars with sample numbers) (Irigoyen, 1997; Irigoyen et al., 1999); the arrow in the Tosca de Mariño section indicates the locality where the section was offset by ~1.5 km. (B) Virtual geomagnetic pole (VGP) latitudes. Solid circles are from class I sites; open circles are from class II sites. (C) Magnetic polarity zonation derived from the VGP latitude plot (B). Black zones have normal polarity; white zones have reversed polarity. Dotted line near the top of section I indicates that site 38 probably spans a magnetic transition because two specimens are reversely magnetized and two specimens from a single block are normally magnetized. Question marks are placed adjacent to polarity zones based on single class II sites to indicate the large uncertainty in their length. Sites are numbered sequentially up the sections beginning with section I. For clarity not all sites are labeled in the figure. Samples at site 04 (section I), collected almost at the contact with the underlying Divisadero Largo Formation, were too weakly magnetized to allow a determination of their magnetization direction.

second component is a viscous remanence (VRM), which is usually directed north and up, roughly parallel to the present Earth's magnetic field. It occupies a blocking temperature (T_b) range below or occasionally overlapping that of the characteristic component.

In sites with a reversed (i.e., south and down) characteristic component, the T_bs of the normal viscous component are readily determined from Zijderveld plots (e.g., Fig. 5, A and D). The distribution of the highest T_bs of the normal-polarity viscous component at all sites with a reversed characteristic component is shown in Figure 6. In most of the samples, the VRM is removed between 100 and 250 °C. In only eight specimens is it possible that the T_b range of the viscous component extends to 350 °C or higher; six of these specimens are from Mariño Formation sites 06, 16, and 22 (for sample locations, see Fig. 3, I). The presence of the VRM at high temperatures in these specimens, however, is somewhat uncertain due to the scattered nature of the results. Two specimens from sites 53 (Fig. 3, II) and 62 (Fig. 4,

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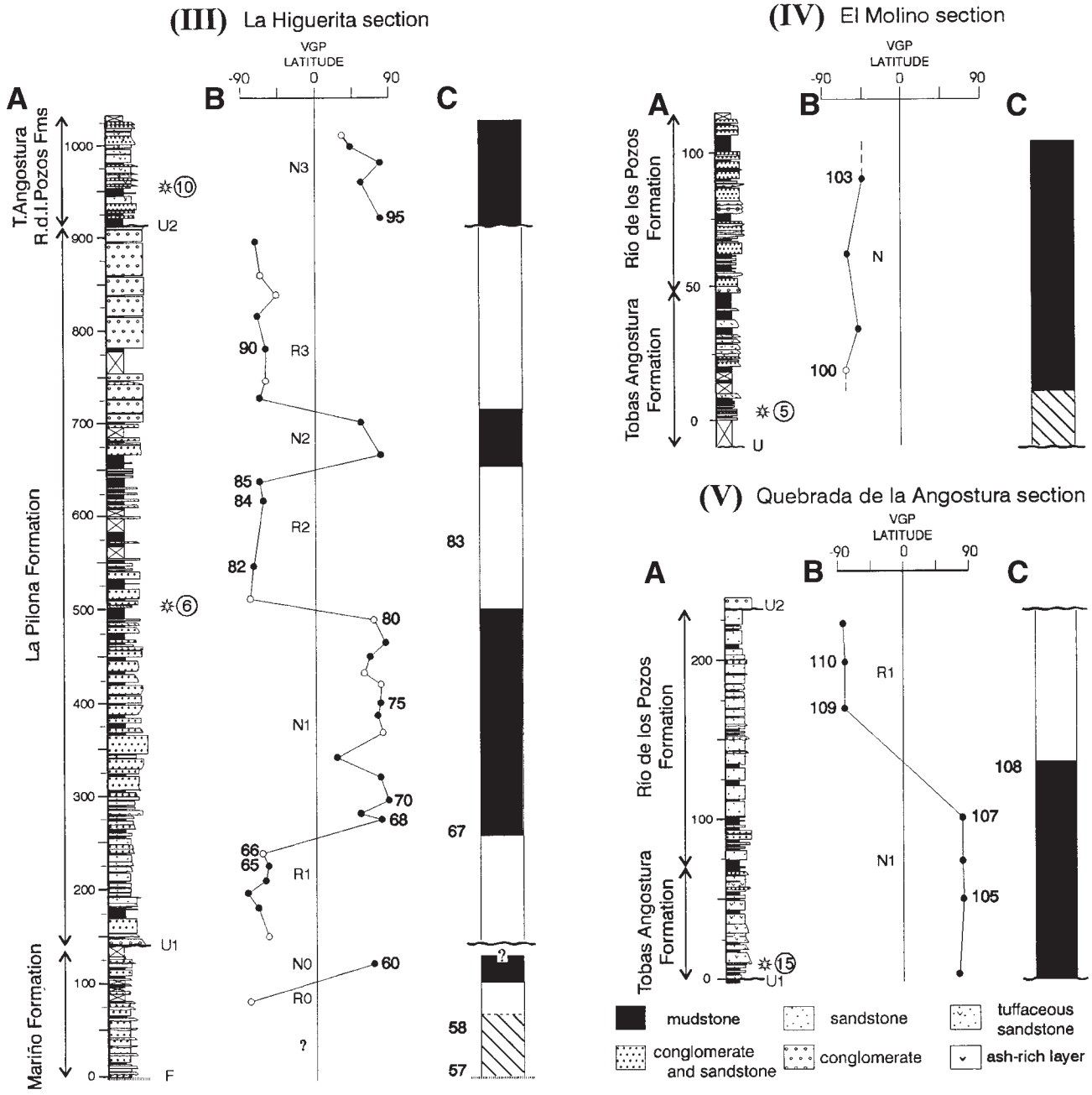


Figure 4. Magnetic polarity stratigraphy of the La Higuerita (III), El Molino (IV), and Quebrada de la Angostura (V) sections. (A) Simplified stratigraphic section, (B) virtual geomagnetic pole (VGP) latitudes, and (C) magnetic polarity zonation as described in Figure 3. Cross-hatched portions in (C) have no magnetic polarity data. The base of the Tobas Angostura Formation in section IV is covered. It has been estimated that there is ~15 m of no stratigraphic coverage, below which there are strata from the La Pilona Formation.

III) of the La Pilona Formation are more stably magnetized and required thermal cleaning to 500 or 530 °C to eliminate the normally magnetized viscous component. An alternative explanation for the anomalously high T_bs of the normal components of these reversely magnetized samples is the coexistence of two stable components of opposite polarity.

As almost all of the well-defined viscous components at reversely magnetized sites have T_bs below 250 °C, it was assumed that normally magnetized sites with T_bs of ≥300 °C carry a normally magnetized primary component above 300 °C. Assuming that normal components with T_bs to 560 °C are viscous would have little effect on the interpretation of polarity stratigraphy in this study, except in the lower part of the Mariño Formation. As most of the normal sites in the lower part of the Mariño Formation have T_bs below 500 °C, the predominance of normal polarity in the lower part of the Mariño Formation could reflect a normal viscous overprint that has not been removed by the demagnetization treatment rather than the primary polarity of the beds. This

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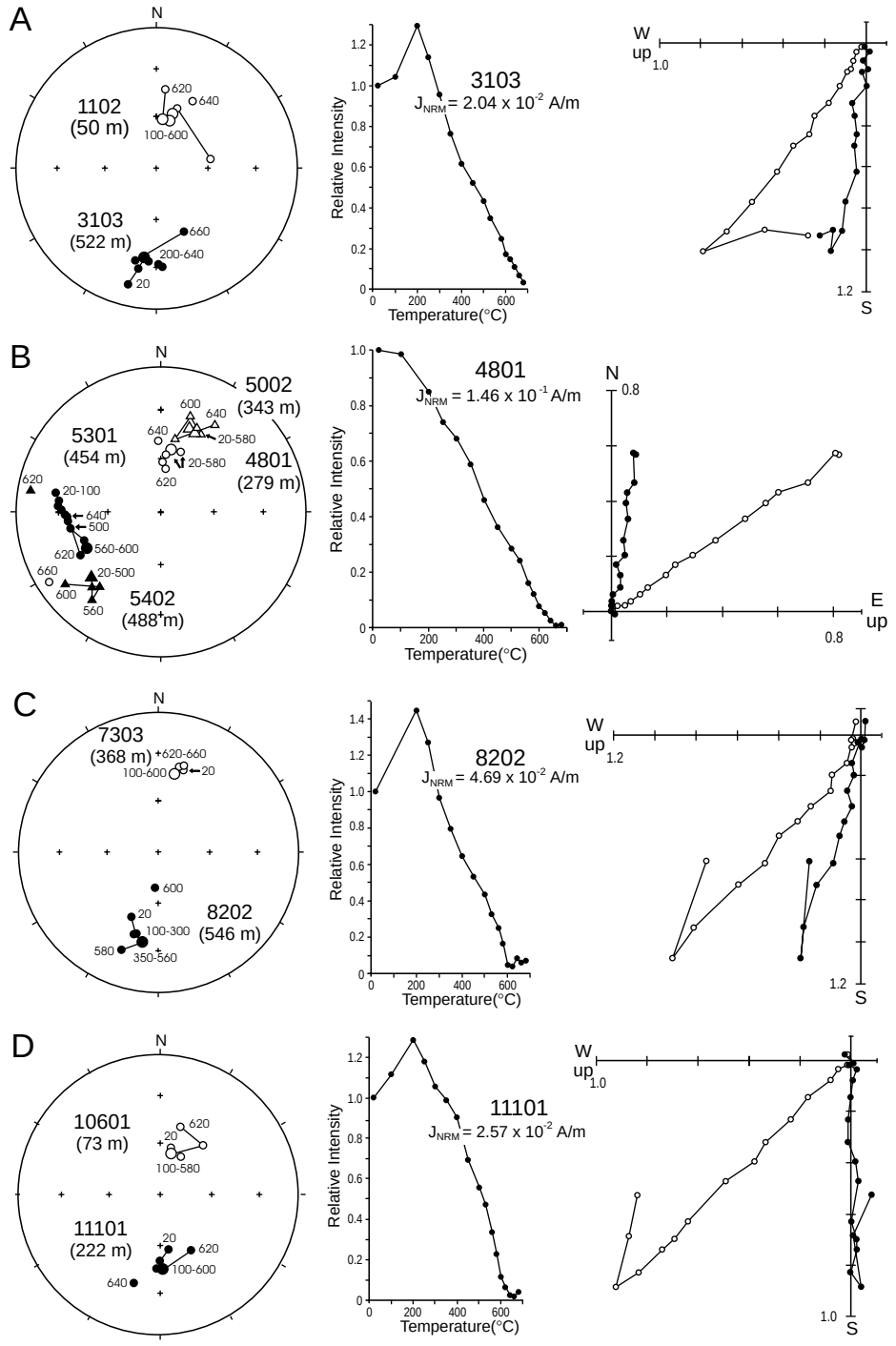


Figure 5. Typical paleomagnetic results for normally and reversely magnetized specimens from the (A) Tosca de Mariño, (B) Arroyo del Agua Blanca, (C) La Higuerita, and (D) Quebrada de la Angostura sections. In the left column, directional data are plotted with respect to paleohorizontal on a Lambert equal-area projection. Open symbols indicate up directions; closed symbols indicate down directions. Random directional data in higher temperature steps are not plotted. In the center column, relative intensity is plotted as a function of temperature. In the right column, Zijderveld plots (Zijderveld, 1967) are shown; closed circles represent projection of the total magnetization vector onto the horizontal plane; open circles indicate projection onto the north-up vertical plane.

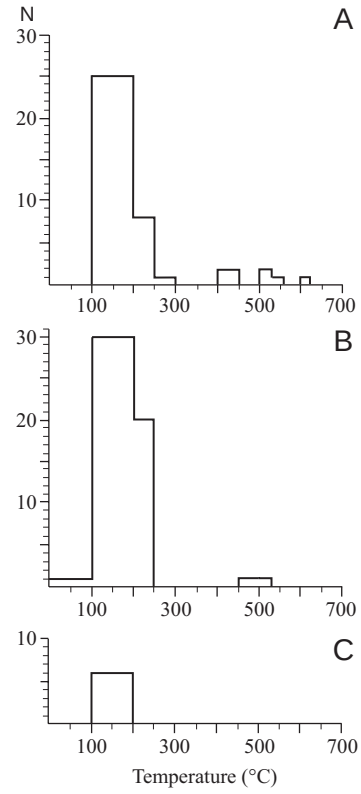


Figure 6. Histograms of the highest blocking temperature of the viscous components at all sites with reversed characteristic component from (A) Mariño, (B) La Pilona, and (C) Río de los Pozos Formations. Viscous blocking temperatures above 400 °C for specimens of the Mariño Formation are rather uncertain, as discussed in the text.

interpretation, although less satisfying, cannot be completely ruled out.

In all formations, the characteristic component, whether of normal or reversed polarity, has a wide range of T_bs. In many samples, T_bs range up to 640–680 °C (e.g., Fig. 5, A and B), demonstrating that hematite is an important carrier of the remanence. In many others, T_bs are below 580–600 °C, suggesting that magnetite may be responsible for the stable remanence (e.g., Fig. 5C). This correlates with earlier work based on isothermal remanent magnetization studies that established the presence of abundant magnetite in rocks of the Mariño Formation (Costa, 1994). In some samples, T_b spectra suggest that magnetite and hematite are carriers of the stable remanence. Magnetic directions at each site are classified according to the system described in the Appendix. We measured the magnetic polarity of 111 sites: 72 sites are designated as class I sites, and 28 sites are class II sites. Mean directions for each class I and II site, after correction for tilting about a hor-

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TABLE 2. MEAN PALEOMAGNETIC RESULTS FOR CLASS I SITES

Section / Polarity	N*	Direction† D (°)	Direction† I (°)	Direction† α_{95} (°)	Direction† k	Paleopole§ Lat (°N)	Paleopole§ Lon (°E)	Paleopole§ dm (°)	Paleopole§ dp (°)
Tosca de Mariño—Arroyo del Agua Blanca									
Normal	24	19.0	–45.0	8.2	14.7	72.2	3.9	10.3	6.5
Reverse	11	210.1	46.7	11.8	15.8	63.7	17.6	15.3	9.8
La Higuerita									
Normal	16	12.6	–47.1	10.9	12.5	78.1	–0.1	14.1	9.1
Reverse	11	201.3	49.1	9.2	25.5	71.6	16.5	12.2	8.1
Quebrada de la Angostura									
Normal	4	2.9	–48.3	10.9	72.1	85.3	324.5	14.3	9.3
Reverse	3	178.0	49.6	15.8	62.2	86.7	258.7	21.0	13.9
El Molino									
Normal	3	341.8	–53.6	19.9	39.4	74.9	192.4	27.8	19.4
Combined data									
Normal	47	13.3	–47.0	5.7	14.3	77.5	1.1	7.4	4.7
Reverse	25	202.6	48.8	6.7	19.9	70.4	17.0	8.8	5.8

*N, number of sites used in statistics.
†D and I, mean declination and inclination, respectively, corrected for tilting of the beds; α_{95}, radius of the circle of 95% confidence about the mean direction; k, precision parameter (Fisher, 1953).
§Latitude (Lat) and longitude (Lon) of the mean paleopole; dm and dp, the semiaxes of the oval of 95% confidence about the mean paleopole.

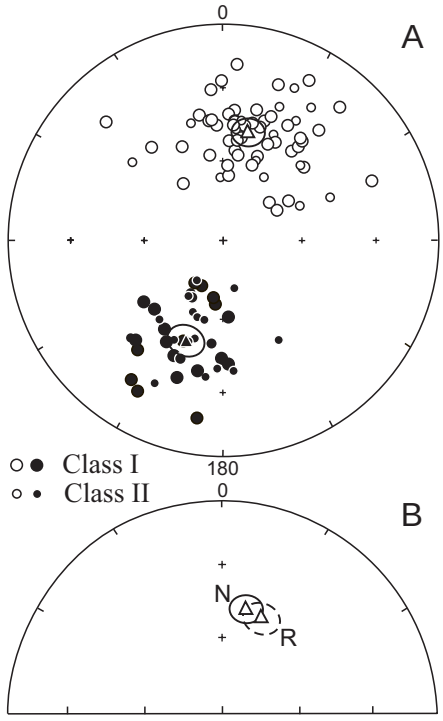


Figure 7. (A) Mean site directions for characteristic component of class I and II sites plotted on a Lambert equal area stereonet with respect to paleohorizontal. Open circles are on upper hemisphere; closed circles are on lower hemisphere. The mean direction of magnetization for normally magnetized class I sites is represented by an open triangle with its circle of 95% confidence; closed triangle represents reversely magnetized sites. (B) Mean directions of normally (N) and reversely (R) magnetized sites with circles of 95% confidence are plotted on the upper hemisphere.

izontal axis, are clustered to the north and up or to the south and down (Fig. 7A). The mean direction for normally magnetized class I sites is D = 13.3°, I = –47.0° (precision parameter, k = 14.6; radius of the 95% confidence circle about the mean direction, α_{95} = 5.7°) with a corresponding paleopole at 77.5°N, 1.1°E (dm = 7.4° and dp = 4.7°, where dm and dp are the semiaxes of the oval of 95% confidence). The mean direction for reversely magnetized class I sites is D = 202.6°, I = 48.8° (k = 19.9, α_{95} = 6.7°), with a corresponding paleopole at 70.4°N, 17.0°E (dm = 8.8°, dp = 5.8°). The normal and reversed means are antipodal (Fig. 7B), supporting the interpretation that these samples are accurate recorders of the Earth's ancient magnetic field and that any later overprint has been removed during the demagnetization procedure.

The mean characteristic directions after tilt correction for normal and reversed class I sites from each paleomagnetic section are listed in Table 2. The magnetization directions from the Tosca de Mariño, Arroyo del Agua Blanca, and La Higuerita sections of the western limb of the La Pilona anticline are rotated clockwise ~13°–30° from the present axial dipole field, which has a zero declination. However, the magnetization direction of the El Molino section on the eastern limb of the anticline is rotated slightly counterclockwise from the present axial dipole field. The observed difference in declination between the two limbs may be related to internal strains in the system that were not accounted for by correcting for tilt about a horizontal axis, or to block rotations about a vertical axis.

Chronology of the Neogene Units

The local magnetic polarity stratigraphies obtained in this study can be correlated with the geomagnetic polarity time scale (GPTS), which in turn is based on marine magnetic anomalies (e.g., Opdyke and Channell, 1996). Several time scales for the Neogene are available. Among the most recent are those of Baksi (1993), Cande and Kent (1995), and Berggren et al. (1995). The GPTS proposed by Baksi (1993) is shown for comparison with the local magnetostratigraphy in Figure 8 and is used throughout the following discussion, because the long normal chron C5n, which has been identified in the studied sections, is precisely dated by the ^{40}Ar-^{39}Ar technique (cf. Baksi et al., 1993). The GPTS of Cande and Kent (1995) and that of Berggren et al. (1995) do not differ from that of Baksi (1993) by more than 0.3 m.y. between 12 and 8 Ma, the interval that includes all the dated tephra beds of this magnetostratigraphic study. However, the Cande and Kent (1995) and Berggren et al. (1995) GPTSs become increasingly discordant with respect to the Baksi (1993) scale for ages greater than 12.0 Ma. Baksi (1993) placed the base of C5ADn at 15.66 Ma, whereas Cande and Kent (1995) and Berggren et al. (1995) located it at 14.61 Ma. This discrepancy stems in part from the choice of different tie points in the middle Miocene.

Tosca de Mariño Section

The Tosca de Mariño section includes strata of the Divisadero Largo (3 sampling sites) and Mariño formations (36 sites) and consists of two separate segments (IA and IB, Fig. 2) located ~1.5 km apart. They share a sandstone unit that is physically continuous from one segment to the other. At the top of the section is a fault that suggests that at least a portion of the Mariño Formation is missing. There are 22 sites from the Mariño Formation that are class I; 9 sites from the Mariño Formation and all 3 sites from the Divisadero Largo Formation are class II. The stratigraphy, location of paleomagnetic sites, and virtual geomagnetic pole (VGP) latitudes used to calculate the magnetic polarity zonation are summarized in Figure 3-I. Examples of stable normally and reversely magnetized specimens from the Mariño Formation are shown in Figure 5A.

The magnetostratigraphy of this section is notable for the long dominantly normal polarity zone (sites 07–19) near the base of the Mariño

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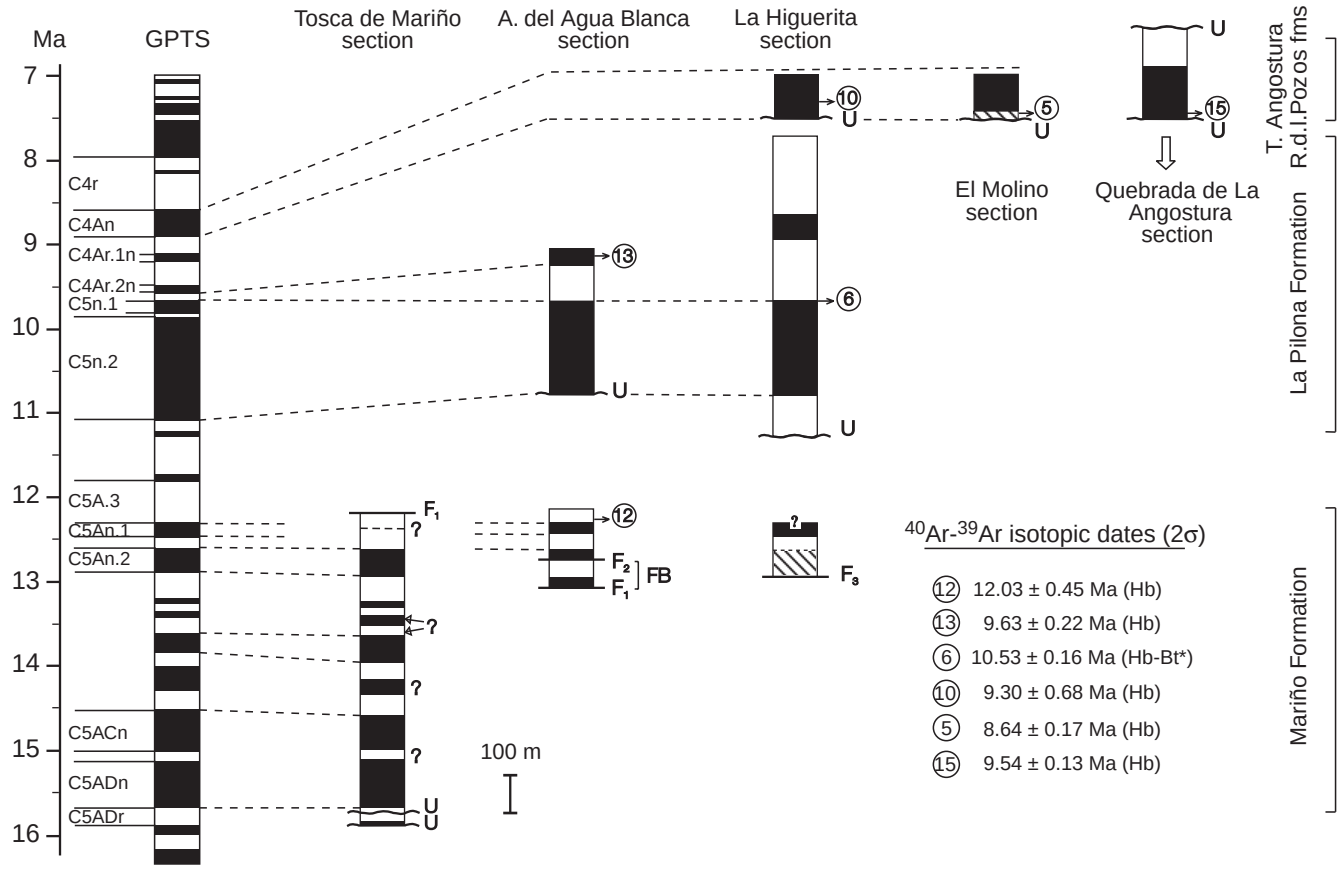


Figure 8. Correlation of the magnetostratigraphy of the Neogene strata of the Mendoza Precordillera with the geomagnetic polarity time scale (GPTS; Baksi, 1993). Black zones represent normal polarity; white zones represent reversed polarity. Cross-hatched portions have no magnetic polarity data. Dashed lines correlate the boundaries of polarity zones among the sampled sections and with the GPTS as discussed in the text. Strata from the fault block (FB) in the lower part of the Arroyo del Agua Blanca section may be placed at a similar stratigraphic level as strata above fault F_2, based on lithostratigraphic correlation. The arrow below the Quebrada de la Angostura section implies that the base of this section, which is interpreted to be the base of the Tobas Angostura Formation, may be as old as ca. 9.5 Ma (see text for further discussion). The ^{40}Ar-^{39}Ar isotopic dates listed in the figure represent the preferred age as established by Irigoyen (1997) and/or Irigoyen et al. (1999); asterisk indicates weighted average age. Sample polarity chrons mentioned in the text are shown to the left of the GPTS; nomenclature is based on Cande and Kent (1995) and Baksi (1993).

Formation and for the large number of reversals encountered. At least 15 geomagnetic reversals spanning a 772 m stratigraphic section are recorded by the Mariño Formation at the sampling locality. Many of the polarity intervals stratigraphically above site 19 are based on data from single sites and their lengths cannot be accurately determined. Nevertheless, a series of short polarity zones is recorded.

The magnetic polarity stratigraphy of the Tosca de Mariño section is rather difficult to interpret above site 23 because of the short polarity intervals that are recorded. The section can perhaps most easily be matched with the Baksi (1993) GPTS between 15.8 and ca. 12.0 Ma (Fig. 8), with the dominantly normal interval between about 15.7 and 14.5 Ma in the marine record (C5ADn-C5ACn) corresponding to the long dominantly normal interval near the base of the sampled section. The series of short polarity intervals in the top two-thirds of the section correspond with the series of short polarities above C5ACn in the GPTS, although a correlation of specific polarity zones is difficult to achieve. A tentative correlation is shown in Figure 8, with the upper normal interval of the section equivalent to C5An.2 or C5An.1. Support for this correlation comes from the isotopic age of 12.03 ± 0.45 Ma (Irigoyen, 1997; Irigoyen et al., 1999) near the top of the Mariño Formation in the nearby Arroyo del Agua Blanca section, which is thought to be lithologically equivalent to the upper portion of the Mariño Formation of the Tosca de Mariño section. In addition, the long normal interval of the sampled section (15.7–14.5 Ma) corresponds approximately to a prominent eolian unit (middle member) that has been regionally correlated with the ca. 20 Ma and 15–14 Ma eolian sandstones that occur north of the study area (Milana et al., 1993; Jordan et al., 1993a).

Arroyo del Agua Blanca Section

The Arroyo del Agua Blanca section (II, Fig. 2) includes the upper part of the Mariño Formation (6 sampling sites) and a portion of the La Pilona Formation (11 sites). At the bottom of the section is a fault block comprising strata of the Mariño Formation, bounded at the bottom by fault F_1 with strike-slip and reverse displacements and at the top by a nearly vertical fault, F_2. A regional unconformity, locally a paraconformity, separates the two formations. Although this unconformity was described in earlier studies (e.g., Yrigoyen, 1993a, and references therein), it is not obvious in the stratigraphic section of this study. A stratigraphic omission is implied by the magnetostrati-

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graphic and geochronological data discussed in the following. Five sites in the Mariño Formation and eight sites in the La Pilona Formation are class I. One site in the Mariño Formation and two sites in the La Pilona Formation are class II. The magnetic polarity stratigraphy, sampling locations, and associated data for this section are shown in Figure 3-II. Examples of paleomagnetic results for individual specimens of the La Pilona Formation are shown in Figure 5B.

The magnetic polarity zonation log derived from the VGP latitudes shows that the ~70 m thick faulted block of the Mariño Formation strata at the bottom of the profile records a normal magnetozone followed by a reversed magnetozone. Above the faulted block, the Mariño strata record two intervals of normal and reversed polarity. The La Pilona Formation has a thick normal magnetozone from about 200 to 438 m followed by reverse and normal polarity zones.

The Arroyo del Agua Blanca section is notable for two reasons: (1) it captures a long normal polarity magnetozone at the base of the La Pilona Formation, and (2) it contains two isotopically dated ash horizons. The ash horizon in the La Pilona Formation yields an age of 9.63 ± 0.22 Ma (Md-13), whereas the ash horizon at the top of the Mariño Formation is dated as 12.03 ± 0.45 Ma (Md-12) (Irigoyen, 1997; Irigoyen et al., 1999). Based on these age constraints and the configuration of the local magnetostratigraphy, the long normal polarity zone is readily correlated with the long normal chron C5n (anomaly 5) of the GPTS, between 11.08 and 9.67 Ma (Fig. 8).

The Md-13 horizon, dated at 9.63 ± 0.22 Ma, falls within a normally magnetized interval that is stratigraphically above the long normal magnetozone. It may be correlated with either chron C4Ar.2n, between 9.57 and 9.50 Ma, or chron C5n.1n, between 9.83 and 9.67 Ma (Baksi, 1993). If correlated with chron C4Ar.2n, then the tiny reversed polarity zone (C5n.1r) near the top of the long normal chron C5n of the GPTS was not sampled in the local section.

The short section of the Mariño Formation at the base of the Arroyo del Agua Blanca profile is interpreted to be bracketed by the interval ca. 12.7–12.0 Ma. Near the top of the section, sample Md-12 yielded an age of 12.03 ± 0.45 Ma. This suggests that the two normal polarity magnetozones between fault F_2 and the dated ash horizon correspond to C5An.2 and C5An.1 of the GPTS. As noted earlier, this correlation and the lithological similarity between the Mariño Formation of the Arroyo del Agua Blanca section and the upper portion of the Mariño Formation in the Tosca de Mariño section suggest that the top of the Mariño Formation in the Tosca de Mariño section is also ca. 12.0 Ma.

If the long normal interval at the bottom of the La Pilona Formation in the Arroyo del Agua Blanca section is correlated with C5n of the GPTS, and if Md-12 is accepted, then there is a gap of at least 1 m.y. in the Arroyo del Agua Blanca section. Based on this interpretation, this study locates the unconformity at the first appearance of a conglomerate bed having characteristics typical of those of the La Pilona Formation.

La Higuerita Section

The La Higuerita section (section III, Fig. 2) includes three formational units, the upper part of the Mariño Formation (4 sampling sites), the La Pilona Formation (34 sites), and the Tobas Angostura Formation (5 sites). The precise locations of the unconformities that separate these three units (cf. Yrigoyen, 1993a) are not obvious in the study section. Because of poor exposure, the location of the contact between the Mariño and the La Pilona Formations was placed arbitrarily at the first appearance of a gray conglomerate with sandstone layers, which is typical of this unit. For the upper unconformity, this study assumed that the base of the Tobas Angostura unit is marked by a tuffaceous greenish mudstone interval with ash-rich sandstone lenses and scattered lapilli. The upper 10 m of this section represents the gradual transition between the Tobas Angostura and Río de los Pozos Formations. Twenty-seven sites in the section are class I sites and 12 sites are class II. The stratigraphy, location of the paleomagnetic sites, and paleomagnetic data from the La Higuerita section are summarized in Figure 4-III. Examples of paleomagnetic results for individual specimens of the La Pilona Formation are shown in Figure 5C.

No information on magnetic polarity of the Mariño Formation was obtained in the lowest 67 m, as the first two sites are class III. Above these sites, the Mariño Formation records a reversed and a normal magnetozone. The La Pilona Formation records a thick normal magnetozone that extends from 264 to 500 m. The remainder of the formation is reversely polarized except for a short normal magnetozone between 653 and 715 m. The ~120-m-thick Tobas Angostura Formation records consistent normal polarity.

Stepwise thermal demagnetization demonstrates that in most La Pilona samples the T_b spectrum of the characteristic component is close to the Curie temperature of hematite. However, samples from the lower part of the La Pilona Formation (sites 63–67) and a few from higher stratigraphic levels (sites 80 and 82) have T_bs no higher than 580–600 °C, suggesting that magnetite may be the major carrier of the stable remanence (e.g., Fig. 5C).

The La Higuerita section is notable for (1) a long normal polarity magnetozone in the La Pilona Formation, and (2) two isotopically dated ash horizons. The Md-6 ash horizon of the La Pilona Formation yields a weighted average age of 10.53 ± 0.16 Ma, and the Md-10 tuff horizon of the Tobas Angostura unit yields an age of 9.30 ± 0.68 Ma (Irigoyen, 1997; Irigoyen et al., 1999). The long normal polarity magnetozone in the La Pilona Formation is assigned to chron C5n (11.08–9.67 Ma) based on its distinctive length and the constraints provided by the isotopic ages. However, although the isotopic ages clearly indicate that the long normal interval is equivalent to chron C5n, the Md-6 isotopic age of 10.53 ± 0.16 Ma at the top of the interval is significantly older than the age assigned to the younger boundary of chron C5n, using either the Baksi (1993) or Berggren et al. (1995) time scale. The younger limit of chron C5n in the Baksi (1993) GPTS is 9.67 ± 0.22 Ma (2σ) based on ^{40}Ar-^{39}Ar plateau ages from three lava flows in the transition zone (Baksi et al., 1993; Baksi, 1993). The age discrepancy could reflect an unidentified erosional unconformity in the study section ca. 9.7 Ma, perhaps within the fine-grained interval of the section, where it would not be readily recognized. If so, the upper portion of the long normal magnetozone would not be recorded and the 10.53 ± 0.16 Ma age could accurately record the age of the ash horizon.

The remainder of the La Pilona Formation above the long normal interval is assigned to reversed chron C4Ar. The age of the normal magnetozone within the long reversed interval near the top of the La Pilona Formation is ambiguous and represents either chron C4Ar.1n or C4Ar.2n. There are no isotopic ages to constrain the age of the Mariño strata below the unconformity that separates this unit from La Pilona Formation. Tentatively, the normal polarity magnetozone recorded in these strata is correlated with chron C5An.1 (12.30–12.48 Ma), based on lithostratigraphic correlation of this part of the section with the exposures in the Arroyo del Agua Blanca section.

The normal polarity magnetozone located above the unconformity that separates the Tobas Angostura and La Pilona Formations is assigned to chron C4An (8.59–8.92 Ma). This correlation is based on (1) the isotopic age of sample Md-10 (Irigoyen, 1997; Irigoyen et al., 1999) and (2) the fact that chron C4An is the first relatively long normal interval above the long reversed chron C4Ar in the GPTS.

El Molino and Quebrada de la Angostura Sections

The short El Molino section (IV, Fig. 2) includes the Tobas Angostura and the Río de los Pozos Formations. The latter unit could not be

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sampled in the La Higuerita section due to the lack of exposure. In view of the gradational nature of the contact between the two formations, the first appearance of a conglomerate with pebbles and cobbles is defined by convention as the boundary between them. Approximately 15 m of strata at the base of the Tobas Angostura Formation are covered and could not be sampled. The stratigraphy, location of paleomagnetic sites, and paleomagnetic data from the El Molino section are summarized in Figure 4-IV. Of four sampling sites, three qualify as class I and one as class II. Paleomagnetic data from all four sites yield a normal polarity magnetozone.

The Quebrada de la Angostura section (V, Fig. 2) covers the Tobas Angostura and Río de los Pozos Formations. In this profile, a distinctive tuffaceous, carbonate-rich dark bed is taken arbitrarily as the boundary between these two units. Of the eight sampling sites in the section, seven are class I. The stratigraphy, location of paleomagnetic sites, and paleomagnetic data from the Quebrada de la Angostura section are summarized in Figure 4-V. The lowest 133 m of the section composing the Tobas Angostura unit and the lower part of Río de los Pozos Formation records a normal magnetozone, whereas the remainder of the section is reversely magnetized. Examples of stable normally and reversely magnetized specimens are shown in Figure 5D.

The El Molino and Quebrada de la Angostura sections include strata of the same units, the Tobas Angostura Formation and the lower part of the Río de los Pozos Formation. The normal polarity magnetozone in both sections is assigned to chron C4An, and the reversed polarity zone in the Quebrada de la Angostura section is correlated with chron C4r. The grounds for this interpretation are (1) the ^{40}Ar-^{39}Ar isotopic ages from both sections, (2) the isotopic age obtained at an equivalent stratigraphic position in the La Higuerita section, and (3) the consistent normal polarity observed in the Tobas Angostura in all three sections.

In the El Molino section, the isotopically dated ash horizon (Md-5) collected ~15 m from the base of the unit yields an age of 8.64 ± 0.17 Ma, in agreement with the age of 9.30 ± 0.68 Ma for sample Md-10 in the La Higuerita section. Therefore, the normal intervals recorded in these two sections are likely correlated with chron C4An of the GPTS. The beginning of this chron is placed at 8.92 Ma based on interpolation between two calibration points at 1.78 and 9.67 Ma (Baksi, 1993). The normal polarity zone in the Quebrada de la Angostura section located ~25 km south of the other two sections yields an age of 9.54 ± 0.13 Ma (Md-15), ~0.5 m.y. older than the base of chron C4An in the GPTS. If this isotopic age is correct, then strata from the Quebrada de la Angostura section record somewhat older stratigraphic levels than those in the La Pilona area, with the boundary between the Tobas Angostura and Río de los Pozos formations occurring at about 9.3 Ma.

Summary

This magnetostratigraphic study of Neogene nonmarine strata of the Mendoza Precordillera suggests the following. (1) Deposition of the Mariño Formation started ca. 15.7 Ma (GPTS of Baksi, 1993) if the proposed correlation between the long normal interval at the base of the Tosca de Mariño section and polarity chron C5ADn-C5ACn is correct. Deposition continued to ca. 12.0 Ma, the age of an ash-fall horizon from the uppermost part of the unit. (2) Deposition of the La Pilona Formation in the La Higuerita section started ca. 11.7 Ma, assuming a constant sedimentation rate for the lower part of this unit, and continued to ca. 9.0 Ma. However, in the Arroyo del Agua Blanca section, deposition of the La Pilona Formation started no earlier than ca. 11.1 Ma, assuming that the unconformity between the Mariño and La Pilona Formations is marked by the first appearance of conglomerate beds typical of the La Pilona Formation. (3) In the La Pilona area, deposition of the Tobas Angostura Formation occurred between ca. 8.9 and 8.7 Ma, and was followed immediately by deposition of the Río de los Pozos Formation. (4) Farther south, in the Tupungato area, isotopic dating suggests that deposition of the Tobas Angostura Formation may have commenced ca. 9.5 Ma, earlier than in the La Pilona area. (5) The stratigraphy of the Neogene succession exhibits regional unconformities that mark hiatuses of ~100 k.y. to ~4 m.y. The unconformity between the Mariño and La Pilona Formations likely represents an interval of ~1.0 m.y. in the Arroyo del Agua Blanca section, but may span a shorter interval in the La Higuerita section. The unconformity between the La Pilona and Tobas Angostura Formations recorded in the La Higuerita section probably represents a short interval of 10–100 k.y. Yrigoyen (1993a) and Irigoyen (1997) suggested that a hiatus of ~3–4 m.y. separates the Río de los Pozos and Mogotes Formations.

TECTONIC IMPLICATIONS

The magnetostratigraphy reported in this paper, in conjunction with ^{40}Ar-^{39}Ar data (Irigoyen, 1997; Irigoyen et al., 1999), establishes age constraints for the syntectonic strata of the Mendoza Precordillera. In addition, this study also establishes a correlation between the sedimentary record and episodes of deformation and/or volcanic activity in the western thrust belts (i.e., the Principal and Frontal Cordilleras) and constrain the timing of deformation of the Neogene succession.

The oldest Mariño strata record the existence of a foreland basin that developed prior to or ca. 16 Ma (GPTS of Baksi, 1993) as a result of a tectonically induced subsidence. The commencement of Neogene sediment accumulation in the study area is coeval with a major phase of thrusting in the Aconcagua fold-and-thrust belt (cf. Ramos, 1996; Cegarra and Ramos, 1996). The correlation between thrusting in the Principal Cordillera and sedimentation in the adjacent foreland is confirmed by a provenance study carried out on rocks of the Mariño Formation and paleocurrent data (Irigoyen, 1997), which will be reported in detail in another paper.

The beginning of deposition of Neogene strata in the study area ca. 16 Ma is also temporally coincident with the eastward shifting of the subduction-related magmatic foci to the Argentine Principal Cordillera. The volcanic and pyroclastic rocks cropping out in the Aconcagua fold-and-thrust belt (Complejo Volcánico Aconcagua; Ramos et al., 1996) are 15.8–8.5 Ma, whereas in the adjacent Chilean region to the west, the volcanic rocks of the Farellones Formation record an important period of magmatic activity between ca. 20.4 and 16.6 Ma (Rivano et al., 1990; Ramos et al., 1991, 1996). Furthermore, geochemical data from the volcanic rocks in this region suggest that the subduction zone of the Nazca plate began to flatten by ca. 16 Ma, long before magmatism ceased, and that thickening of the continental crust related to the tectonic stacking of the Andean Cordillera evolved since about that time (Kay et al., 1991, and references therein; Ramos et al., 1991).

This study demonstrates that the initiation of foreland-basin subsidence and deposition at the study site are similar in age to the development of thin-skinned tectonics in the central part of the Aconcagua fold-and-thrust belt (e.g., Ramos, 1988, 1996; Cegarra and Ramos, 1996) and, hence, are genetically linked. The initiation of an important phase of thrusting in the Principal Cordillera and the eastward migration of the calc-alkaline volcanic axis may be linked with the flattening of the descending plate ca. 16 Ma (cf. Ramos et al., 1991; Ramos, 1996).

Episodes of volcanic activity occurring in the Cordilleran region were sometimes recorded in the study area, as inferred from the age of the air-fall tephras interbedded in the nonmarine deposits (Irigoyen, 1997; Irigoyen et al., 1999). For example, the earliest 15.8–8.5 Ma volcanic activity in the Aconcagua region is recorded by bentonite horizons of the ca. 15.7 Ma lower member of the Mariño Formation and by the highly tuffaceous nature of the middle member, which contains

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abundant glass shards. In addition, the extensive ash-rich unit, the ca. 8.9–8.7 Ma Tobas Angostura Formation, is correlated with the latest extrusive episode in the Principal Cordillera ca. 8.5 Ma. This latter correlation is supported by the lateral variations in thickness of the Tobas Angostura unit, which tapers to the east-southeast (cf. Yrigoyen et al., 1954), suggesting that the eruptive vents were located to the west-northwest of the study area in the direction of the Aconcagua peak. In addition, the facies assemblage of the Tobas Angostura Formation (cf. Irigoyen, 1997) accounts for a volcanic episode that included a voluminous pyroclastic eruption, which was responsible for providing abundant volcanic detritus (lapilli, ashes) that overwhelmed the basin. This interpretation is consistent with the occurrence of a thick sequence of dominantly proximal pyroclastic-flow deposits, tuffs, and lavas, and with the presence of gas-escape structures in the uppermost part of the Aconcagua mountain (cf. Ramos et al., 1996), and suggests that the latest eruptive episode was likely the largest in the region.

Exhumation and displacement of the Frontal Cordillera overlapped in time with deformation in the Principal Cordillera (Irigoyen, 1997). The latest phase of deformation in the Aconcagua fold-and-thrust belt is thought to have occurred at 9–8 Ma (cf. Ramos, 1996; Cegarra and Ramos, 1996). A multiple data set that includes timing of facies changes, paleocurrent data, accumulation rates, and provenance allows a temporal correlation to be made between deposition of the ca. 11.7–9.0 Ma La Pilona Formation and exhumation of the Frontal Cordillera to the northwest of the study site (Irigoyen, 1997). Immediately east of the Las Carreras fault system, on the eastern flank of the Cordón del Plata (Fig. 1), the youngest Mogotes Formation includes conglomerates with very large boulders of Paleozoic magmatic and sedimentary rocks that were almost certainly derived from the Frontal Cordillera immediately to the west. Therefore, the basal unconformity of the Mogotes Formation is likely the result of exhumation and deformation in the flanking thrust belt and probably developed as a result of diachronous pulses of deformation along the Cordón del Plata (Irigoyen, 1997).

Deformation of Neogene strata occurred in the late Pliocene–early Pleistocene(?), as discussed in the following. The present location of the orogenic front at lat 33°S, in the eastern foothills of the Precordillera (e.g., Figueroa and Ferraris, 1989), clearly indicates that deformation has progressed eastward since deposition of the Mogotes Formation. Deformation is still active, as inferred from earthquake focal mechanisms (Triep, 1987; Ramos, 1990), intense compressive deformation, and evidence of neotectonic activity in the study area (Irigoyen, 1997) and farther east, in the foothills of the Mendoza Precordillera (Bastías et al., 1993).

Combined paleomagnetic and tephra chronologies on the continental deposits are particularly useful in providing temporal constraints for the timing of oil generation and development of prospective closures. The study area encompasses a region of active oil production known as the Cuyo basin, an elongate northwest-striking basin that includes Middle to Upper Triassic nonmarine strata and subordinate volcanic rocks, interpreted to be the sedimentary fill of a continental rift basin (Kokogian et al., 1993, and references therein). Hydrocarbon reserves generated mainly from the Late Triassic Cacheuta Formation are concentrated in several oil fields; one, the Cacheuta-Tupungato depocenter (Kokogian et al., 1993), includes the La Pilona and Tupungato anticlines. The structure of the region based on subsurface data has been published in simplified form (e.g., Sarewitz, 1988; Kozlowski et al., 1993). According to Jordan and Ortiz (1987), the thickness of sedimentary cover above the Cacheuta Formation at the end of deposition of the youngest Mariño strata is about 2500 and 3000 m in the Cacheuta and Potrerillos areas, respectively, sufficient for the beginning of oil generation. This study indicates that these conditions were not achieved before ca. 12 Ma, the age of the youngest Mariño strata, and were enhanced after deposition of the overlying ~1000-m-thick La Pilona Formation, the youngest strata of which are dated as ca. 9 Ma.

Structural traps developed during regional east-west contraction in the late Pliocene–early Pleistocene(?), and most of the closures relate to folds and reverse faults genetically linked with the reverse reactivation of Triassic major normal faults (Dellapé and Hegedus, 1993, and references therein). Motion on the La Pilona and Tupungato structures started after the beginning of deposition of the Mogotes Formation ca. 3.0 Ma (Yrigoyen, 1993a); and ceased prior to ca. 0.4–0.3 Ma, the ages of the air-fall horizons interbedded with Quaternary gravels in the La Pilona and Tupungato areas (Irigoyen, 1997; Irigoyen et al., 1999). The occurrence of growth geometries in strata of the Mogotes Formation cropping out ~35 km farther east, in the Barrancas anticline (Chiaramonte, 1996), suggests that deformation may have been also partially synchronous with deposition of this unit in the study area. Fracturing of potential reservoir rocks (Potrerillos, Río Blanco, Barrancas, and Papagayo Formations of Triassic-Paleocene age; Dellapé and Hegedus, 1993) may have occurred during development of closures, improving their permeability and providing conduits for oil under pressure to migrate along fracture planes into previously sealed reservoir-quality rocks.

Magnetostratigraphy also allows calibration of the ages of land mammalian fossils collected in the Neogene units of the study area, enhancing the precision of biostratigraphic ages. In particular, the Río de los Pozos Formation provided abundant fossil remains that were collected in the eastern flank of the Tupungato anticline, <2 km to the northeast of the Quebrada de la Angostura section (see Fig. 1 in Yrigoyen, 1993b). Previous studies assigned the Río de los Pozos Formation to the Pliocene (e.g., Pascual and Odreman Rivas, 1973) or to the latest Miocene–early Pliocene (e.g., Yrigoyen, 1993a), based on the fauna described in the following. As summarized by Yrigoyen (1993b) and listed in Pascual and De la Fuente (1993), the faunal assemblage includes the Hydrochoeridae *Kiyutherium* aff. *K. orientalis* Francis and Mones of the order Rodentia, the Mesotheriidae *Typotheriopsis silverai* Cabrera and the Hegetotheriidae *Hemihegetotherium achathaleptum* Rovereto, both of the order Notoungulata, and a cranium of a capybara-type animal of the subfamily Cardiatheriinae *(Procardiotherium* sp.?). This assemblage was collected near the ca. 9.3 Ma contact between the Río de los Pozos Formation and the underlying Tobas Angostura unit. Therefore, the mammal-bearing strata are late Miocene (Tortonian) rather than Pliocene and hence contain Huayquerian fauna instead of a Pliocene Montehermosan fauna according to land-mammal age boundaries defined by Pascual and De la Fuente (1993). Indeed, the Huayquerian land mammal age (Simpson, 1940) is named for the Huayquerías Formation that crops out in the area ~100 km south of the city of Mendoza, in which vertebrate remains of late Miocene age occur (Marshall et al., 1986).

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APPENDIX

^{40}Ar-^{39}Ar Geochronologic Laboratory Procedures of Irigoyen et al. (1999)

Pure mineral separates of hornblende and biotite were prepared using standard heavy liquid and magnetic methods followed by handpicking. Mineral irradiation was performed in the research reactor at McMaster University, Hamilton, Ontario. Mineral separates were analyzed at the Massachusetts Institute of Technology using resistance furnace step-heating and laser total fusion techniques, and at the Geological Survey of Canada using the laser ^{40}Ar-^{39}Ar step-heating method. Procedures and instruments are similar to those explained in Hodges et al. (1994) and Villeneuve and MacIntyre (1997), respectively. Ages for Massachusetts Institute of Technology and Geological Survey of Canada analyses are referenced to Fish Canyon Tuff biotite (FCT-3) and Fish Canyon Tuff sanidine (FCT-Kf), respectively. In both cases, apparent ages are assumed to be 28.03 ± 0.1 Ma (Renne et al., 1994). More details were given in Irigoyen (1997) and Irigoyen et al. (1999).

Paleomagnetic Laboratory and Analytical Procedures

In the laboratory, hand samples were cut and ground into 2.5 cm cubes. Measurements were carried out in a magnetically shielded room (magnetic field <3000 nT) at the Geological Survey of Canada in Ottawa. Specimen remanence was measured with a DSM-1 digital spinner magnetometer. Demagnetization experiments on 10 pilot specimens were conducted to determine whether thermal or alternating magnetic field (AF) demagnetization provided the most effective method of isolating the stable component of magnetization. Specimens were AF demagnetized in 15 to 20 steps to a peak field of 100 mT using a Schonstedt single axis demagnetizer, model GSD-1. AF demagnetization had little effect on the magnetization on some of the pilot specimens.

Following the pilot study, systematic stepwise thermal demagnetization experiments were carried out on the rest of the collection. For most sites, at least one specimen was demagnetized in 16 steps to peak temperatures of 100, 200, 250, 300, 350, 400, 450, 500, 530, 560, 580, 600, 620, 640, 660, and 680 °C using a Schonstedt thermal demagnetizer, model TSD-1, modified to maintain temperatures within ±3 °C. If the first specimen that was measured from a given site revealed a stable component of magnetization, the second specimen was subjected to fewer heating steps, usually 250, 350, 450, 500, 530, 550, and 630 °C. Additional steps were included when appropriate. If the first specimen displayed a complex pattern of demagnetization, the 16 step heating procedure was also used for the second specimen. The third specimen of stable sites was demagnetized in two or three steps, generally 500, 550, and 600 °C. Occasionally, more steps were included.

For most sites, magnetization directions for individual specimens were established using the least-squares fit method of Kirschvink (1980). In a few specimens, the magnetization directions of the characteristic component were too scattered to employ this method, and an end point was calculated as the mean of the semistable directions using Fisher statistics (Fisher, 1953). Mean directions for each site, before and after correction for tilting about a horizontal axis, were calculated using Fisher statistics. The corresponding virtual geomagnetic poles (VGPs) were calculated and divided into three classes. Class I sites have a mean magnetic direction based on three samples and satisfy Watson's (1956) criterion for nonrandomness at the 95% confidence level ($R \geq 2.62$). Class II sites are those with $N = 2$ and those with $N = 3$ and $R < 2.62$. For class II sites the polarity is not in doubt. The length of magnetic polarity zones based on single class II sites is poorly constrained. Class III designates those sites where the three specimens have widely divergent directions. Class III sites are not used to define polarity zones. They are often located between sites of opposite polarity and may, in some cases, represent transitions between the two polarities. The boundary between zones of different polarity is placed midway between consecutive normal and reversed data points.

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